we knew how to kill the mosquitoes; but if a thought buzzed in our mind and kept us awake at night, most of us did not know how to kill the thought.

The revolutions in biotech and infotech will give us control of the world inside us, and will enable us to engineer and manufacture life. We will learn how to design brains, extend lives, and kill thoughts at our discretion. Nobody knows what the consequences will be. Humans were always far better at inventing tools than using them wisely. It is easier to manipulate a river by building a dam across it than it is to predict all the complex consequences this will have for the wider ecological system. Similarly, it will be easier to redirect the flow of our minds than to divine what it will do to our personal psychology or to our social systems.

In the past, we have gained the power to manipulate the world around us and to reshape the entire planet, but because we didn't understand the complexity of the global ecology, the changes we made inadvertently disrupted the entire ecological system and now we face an ecological collapse. In the coming century biotech and infotech will give us the power to manipulate the world inside us and reshape ourselves, but because we don't understand the complexity of our own minds, the changes we will make might upset our mental system to such an extent that it too might break down.

The revolutions in biotech and infotech are made by engineers, entrepreneurs and scientists who are hardly aware of the political implications of their decisions, and who certainly don't represent anyone. Can parliaments and parties take matters into their own hands? At present, it does not seem so. Technological disruption is not even a leading item on the political agenda. Thus during the 2016 US presidential race, the main reference to disruptive technology concerned Hillary Clinton's email debacle,[3] and despite all the talk about job losses, neither candidate addressed the potential impact of automation. Donald Trump warned voters that the Mexicans and Chinese will take their jobs, and that they should therefore build a wall on the Mexican border.[4] He never warned voters that the algorithms will take their jobs, nor did he suggest building a firewall on the border with California.

This might be one of the reasons (though not the only one) why even voters in the heartlands of the liberal West are losing faith in the liberal story and in the democratic process. Ordinary people may not understand artificial intelligence and biotechnology, but they can sense that the future is passing them by. In 1938 the condition of the common person in the USSR, Germany or the USA may have been grim, but he was constantly told that he was the most important thing in the world, and that he was the future (provided, of course, that he was an 'ordinary person' rather than a Jew or an African). He looked at the propaganda posters – which typically depicted coal miners, steelworkers and housewives in heroic poses – and saw himself there: 'I am in that poster! I am the hero of the future!'[5]